World-Formation Experimental Programme v1.0 A Falsifiable Experimental Programme for Purpose-Bearing, Self-Revising Observers

A Falsifiable Experimental Programme for Purpose-Bearing, Self-Revising Observers

Version 1.0 — 2026

Abstract

The World-Formation Experimental Programme converts the Formal Core into a staged programme of falsifiable experiments.

The programme does not ask whether World-Formation Theory is globally “true.” It asks whether specific proposed relations survive controlled tests. Its methodological rule is:

Do not test the whole theory. Test the arrows.

The initial experimental architecture therefore separates the functional components of world-formation into independently testable modules: Gate, Trace, Filtration, Residual, Latching, Purpose, Revision Attribution, Meta-Declaration, and later, only if justified, deeper geometric structure.

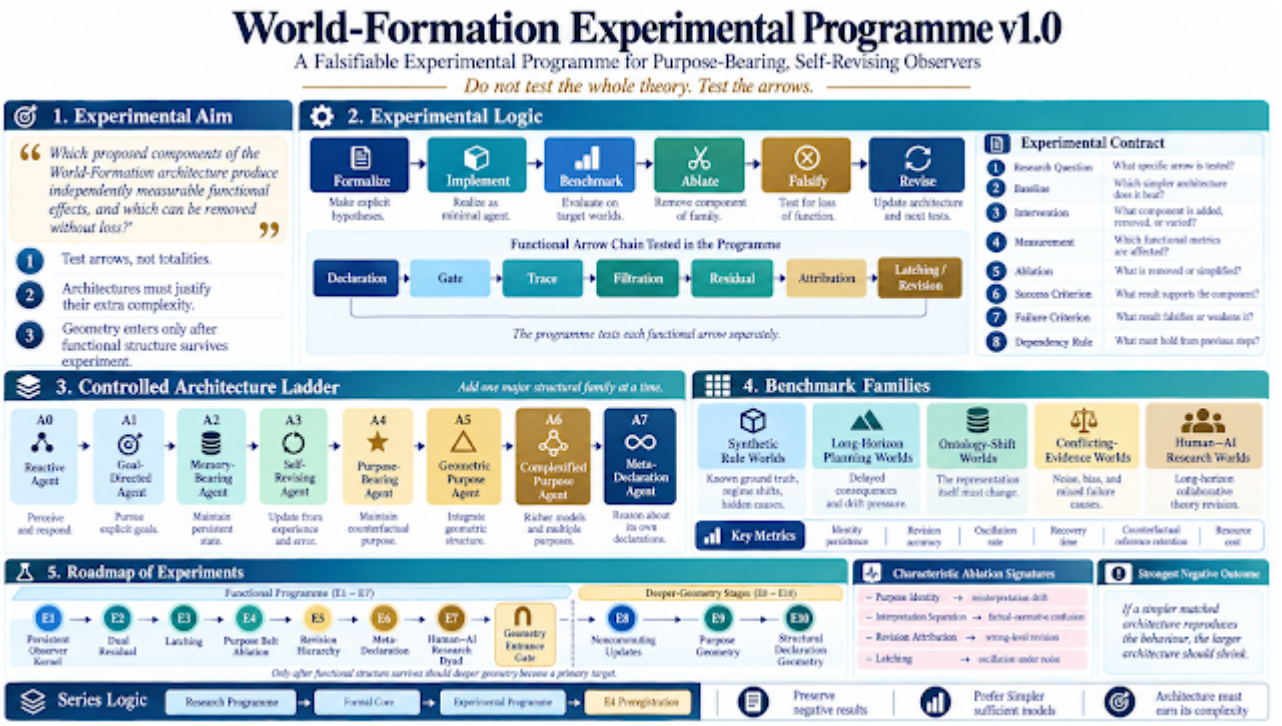
The first experimental phase remains deliberately generic. It does not require octonions, quaternions, complex numbers, symplectic geometry, G2/SO(4), Clifford structure, or any traditional interpretive system. The source development explicitly recommends an AGI ablation ladder beginning with reactive and goal-directed systems, progressing through memory-bearing and self-revising agents, then adding Purpose Belt, geometric Purpose, complexification, and finally Meta-Declaration. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

The programme is organized around four immediate work packages already identified in the source material: Persistent Observer Kernel, Purpose Belt Kernel, Purpose Geometry, and Meta-Declaration / PORE. Each is to be formalized, implemented, benchmarked, ablated, and falsified. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

A major methodological commitment is that architectural complexity must justify itself. A component is not confirmed merely because a larger system performs better. It must either produce a distinctive functional advantage, a characteristic failure mode when removed, a formally irreducible role, or a predictive structure that simpler matched systems cannot reproduce.

The deeper mathematical programme enters only after the functional architecture survives these

tests.



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1. Experimental Objective

The Formal Core proposes the following functional cycle:

Declaration → Gate → Trace → Filtration → Residual → Attribution → Latching / Revision → New Declaration. (1.1)

Purpose supplies a persistent counterfactual reference across this cycle.

The Experimental Programme asks:

> **Which components in this cycle are genuinely necessary, which are useful but optional, and which are merely descriptive re-labellings of mechanisms already available in simpler systems?**

The central operational question is therefore not:

“Does the full architecture work?”

It is:

“Which structural difference causes which measurable difference?” (1.2)

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2. Experimental Contract

Every experiment in the programme must specify eight items before evaluation begins.

2.1 Research Question

What exact arrow is being tested?

For example:

Persistent Purpose → lower reinterpretation drift? (2.1)

Directional Residual → better structural-bias detection? (2.2)

Latching → lower revision oscillation? (2.3)

Revision Attribution → more appropriate revision level? (2.4)

2.2 Baseline

What simpler architecture could plausibly reproduce the proposed effect?

A proposed component must be tested against a strong baseline, not merely against its absence.

2.3 Intervention

What single structural feature differs between treatment and control?

Whenever possible:

ΔArchitecture = one functional component. (2.5)

2.4 Measurement

What observable quantity represents success?

Examples include:

identity persistence,
revision accuracy,
recovery time,
oscillation rate,
false commitment rate,
catastrophic model revision,
counterfactual-reference retention,
resource cost.

The source explicitly states that benchmark evaluation should extend beyond ordinary task score and include long-horizon identity persistence, goal drift, recovery after perturbation, catastrophic model revision, noise versus systematic-bias discrimination, counterfactual reference retention, self-correction without identity collapse, cross-context consistency, appropriate revision-level selection, regime-shift adaptation, and resource cost. 𝕆 → G₂_SO(4) → ℍ → ℂ² 成界過程初探 1...

2.5 Ablation

What happens when the proposed component is removed?

A component without a characteristic ablation effect should not automatically be treated as irreducible.

2.6 Practical Success Criterion

What minimum effect is large enough to justify the additional architecture?

Statistical significance alone is insufficient.

2.7 Failure Criterion

What result would cause the hypothesis to be downgraded, revised, or rejected?

2.8 Dependency Rule

What later experiment becomes justified only if this experiment succeeds?

This prevents downstream mathematics from being introduced before upstream function has been established.

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3. Shared Agent Architecture

A generic experimental state is:

$\Omega_n = (D_n, P_n, I_n, W_n, x_n, F_n, R_n).$ (3.1)

where:

D_n = Declaration,
P_n = Purpose identity,
I_n = Purpose interpretation,
W_n = world model,
x_n = operational state,
F_n = filtration or historical state,
R_n = residual.

A generic episode may be written:

$x_{n+1} = \Phi_D_n,P_n(x_n,u_n,\xi_n).$ (3.2)

$z_n = H_D_n(x_{n+1}).$ (3.3)

$g_n = G(z_n \mid D_n,P_n,I_n,F_n).$ (3.4)

$T_n = \mathrm{Commit}(g_n, z_n).$ (3.5)

$F_{n+1} = F_n \vee T_n.$ (3.6)

$R_{n+1} = E(D_n, P_n, I_n, W_n, F_{n+1}).$ (3.7)

$\ell_{n+1} = A(R_{n+1} \mid D_n, P_n, I_n, W_n, F_{n+1}).$ (3.8)

$\hat{\Omega}_{n+1} = U^{\wedge}(\ell_{n+1})(\Omega_n, R_{n+1}).$ (3.9)

$\Omega_{n+1} = \mathrm{LatchOrAccept}(\Omega_n, \hat{\Omega}_{n+1}; \kappa_\ell).$ (3.10)

Not every experimental condition receives every variable.

The purpose of the programme is precisely to determine which variables are necessary.

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4. Controlled Architecture Ladder

The initial experiments should use a staged ablation ladder.

The source proposes a progression from reactive agents through goal-directed, memory-bearing, self-revising, Purpose-bearing, geometrically enriched, and meta-declarative agents. $\mathbb{O} \to G_2_SO(4) \to \mathbb{H} \to \mathbb{C}^2$ 成界過程初探 1…

A normalized ladder is:

A0 — Reactive Agent. (4.1)

A1 — Goal-Directed Agent. (4.2)

A2 — Memory-Bearing Agent. (4.3)

A3 — Self-Revising Agent. (4.4)

A4 — Purpose-Bearing Self-Revising Agent. (4.5)

A5 — Geometric Purpose Agent. (4.6)

A6 — Complexified Purpose Agent. (4.7)

A7 — Meta-Declaration Agent. (4.8)

Each level should introduce only one major new structural family.

This allows causal interpretation.

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5. Common Benchmark Families

The programme should not rely on one task class.

At least five benchmark families are needed.

5.1 Synthetic Rule Worlds

These environments have explicit ground truth.

Examples:

hidden transition rules,
regime switching,
latent categories,
controlled observational corruption,
delayed consequences.

Their main advantage is that the experimenter knows whether the true failure lies in state, policy, world model, Purpose interpretation, or higher-level Declaration.

5.2 Long-Horizon Planning Worlds

The agent must maintain a stable high-level objective across many local decisions.

These environments test:

goal drift,
shortcut attraction,
history dependence,
Purpose retention,
delayed failure.

5.3 Ontology-Shift Worlds

The representation itself changes.

For example:

$C_1 = \{A,B\}.$ (5.1)

Later:

$C_2 = \{A,B,C\}.$ (5.2)

or:

linear model → nonlinear model. (5.3)

The agent must decide whether to update parameters inside its current model or revise the model class itself.

5.4 Conflicting-Evidence Worlds

Evidence is deliberately heterogeneous.

Some anomalies reflect:

noise,
policy failure,
world-model failure,
Purpose interpretation failure,
genuine structural change.

These environments are essential for Revision Attribution.

5.5 Long-Horizon Research Worlds

An agent conducts extended hypothesis formation, testing, correction, and theory revision.

The source identifies long-horizon Human–AI research as a particularly useful probe because humans repeatedly contribute persistent Purpose, anomaly sensing, reframing, epistemic-status governance, theory genealogy, negative-result interpretation, and judgement about which level should be revised. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

These tasks are therefore especially relevant to Purpose-bearing architectures.

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6. Experiment E1 — Persistent Observer Kernel

6.1 Research Question

Does the minimal observer chain:

Gate → Trace → Filtration → Residual → Latching → Revision (6.1)

produce measurable functional advantages relative to simpler agents?

The source identifies this as the first major work package and explicitly recommends proving or testing which modules are irreducible. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

6.2 Conditions

O0 — Reactive. (6.2)

O1 — Reactive + Memory. (6.3)

O2 — Memory + Gate. (6.4)

O3 — Gate + Explicit Residual. (6.5)

O4 — Residual + Latching. (6.6)

O5 — Full Persistent Observer Kernel. (6.7)

All conditions should be matched for base-model capability and resource budget as closely as possible.

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7. E1 Metrics

7.1 Historical Consistency

HC = 1 – contradictory committed rewrites / committed traces. (7.1)

7.2 False Commitment Rate

FCR = incorrect committed traces / all committed traces. (7.2)

7.3 Revision Precision

RP = correct revisions / all revisions. (7.3)

7.4 Revision Recall

RR = detected true regime changes / all true regime changes. (7.4)

7.5 Revision Oscillation

RO = reversal revisions / all revision events. (7.5)

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8. E1 Ablation Logic

The programme predicts different characteristic failures.

Remove Gate → increased false commitment. (8.1)

Remove Trace → impaired historical consistency. (8.2)

Remove Residual → impaired mismatch detection. (8.3)

Remove Latching → revision oscillation. (8.4)

Remove Revision → persistent model mismatch. (8.5)

If these failures do not separate empirically, the architecture should be compressed.

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9. Experiment E2 — Dual Residual Ledger

9.1 Research Question

Does separating residual magnitude from residual direction improve discrimination between noise and systematic structural bias?

The source explicitly argues that these are different quantities: large residual magnitude may arise from noise, while small but directionally coherent residuals may indicate structural mismatch. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

9.2 Null Hypothesis

A scalar residual magnitude is sufficient.

H_0: Performance($\mathscr{R}$) ≈ Performance($\mathscr{R}$,$\mathcal{G}$). (9.1)

9.3 Alternative Hypothesis

Directional residual provides additional predictive information.

H_1: Performance($\mathscr{R}$,$\mathcal{G}$) > Performance($\mathscr{R}$). (9.2)

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10. E2 Synthetic Construction

Let e_t denote prediction error.

Noise condition

$e_t \sim N(0,\sigma^2)$. (10.1)

Then:

Σe_t^2 may be large. (10.2)

But:

$\Sigma e_t \approx 0$. (10.3)

Structural-bias condition

$e_t = \mu + \varepsilon_t, \mu \neq 0$. (10.4)

Then repeated error may remain directionally coherent.

Compare:

R-only agent = f($\mathscr{R}$). (10.5)

Dual-ledger agent = f($\mathscr{R}$,$\mathcal{G}$). (10.6)

Primary outcome:

false structural revision under noise vs missed structural revision under bias. (10.7)

If dual residual produces no additional discrimination, directional residual remains optional.

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11. Experiment E3 — Latching

11.1 Research Question

Does finite revision resistance improve stability without producing excessive rigidity?

A generic revision rule is:

revise if $\Delta L > \kappa$. (11.1)

otherwise:

Latch. (11.2)

The source explicitly proposes this finite regime between unrestricted adaptation and frozen identity. $\mathbb{O} \to G_2_SO(4) \to \mathbb{H} \to \mathbb{C}^2$ 成界過程初探 1...

11.2 Parameter Sweep

Evaluate:

$\kappa \in \{0, \kappa_1, \kappa_2, \ldots, \kappa_m\}$. (11.3)

Measure:

false revisions,
missed revisions,
adaptation delay,
oscillation,
cumulative task loss.

11.3 Working Hypothesis

A useful regime should exhibit:

$\kappa^* > 0$. (11.4)

but:

$\kappa^* < \infty$. (11.5)

The expected trade-off is:

stability ↔ adaptability. (11.6)

If κ = 0 is consistently optimal, latching loses support as a general primitive.

If very large κ is consistently optimal, the system may not require genuine revision.

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12. Experiment E4 — Purpose Belt Ablation

E4 is the first major confirmatory experiment in the programme.

The source directly proposes:

Self-revision – Purpose Belt (12.1)

versus:

Self-revision + Purpose Belt. (12.2)

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The target is not short-task accuracy.

The source explicitly narrows the strongest regime to:

ontology shift + long horizon + value ambiguity + conflicting evidence + self-revision. (12.3)

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13. E4 Purpose Components

The minimal Purpose architecture separates:

Purpose Identity, (13.1)

Purpose Interpretation, (13.2)

World Model, (13.3)

Realised History, (13.4)

Revision Attribution, (13.5)

Hierarchical Latching. (13.6)

The source further argues that the observer already supplies Trace and Filtration, so the Purpose architecture should reuse these rather than introduce a duplicate memory system. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

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14. E4 Core Ablations

Four decisive ablations are already specified in the source.

Remove Purpose Identity. (14.1)

Predicted failure:

long-horizon reinterpretation drift. (14.2)

Merge Purpose Interpretation into ordinary world-model state. (14.3)

Predicted failure:

factual–normative confusion. (14.4)

Remove Revision Attribution. (14.5)

Predicted failure:

wrong-level revision. (14.6)

Remove Latching. (14.7)

Predicted failure:

oscillation or drift under noisy or adversarial evidence. (14.8)

The source explicitly states that if none of these removals causes meaningful failure, the Purpose Belt has not justified itself. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

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15. E4 Strong Failure Condition

The strongest Purpose claim fails if a simpler representation reproduces both action and revision behaviour.

Formally, if a simpler state Z satisfies:

$P(A_{t:t+k}, U_{t:t+k} \mid Z) \approx P(A_{t:t+k}, U_{t:t+k} \mid B_Purpose)$, (15.1)

across ontology shifts and long-horizon tasks, then:

Purpose Belt irreducibility is not established. (15.2)

The source explicitly identifies this as the key minimality test and strongest failure condition. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

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16. Experiment E5 — Revision Hierarchy

16.1 Research Question

Can different discrepancy classes be mapped to different optimal revision levels?

Let:

$\ell \in \{\pi, W, I, P, D\}$. (16.1)

where:

π = policy,
W = world model,
I = Purpose interpretation,
P = Purpose identity,
D = structural Declaration.

The source explicitly argues that knowing discrepancy exists is insufficient; the agent must determine what kind of thing is wrong. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

16.2 Environment Design

Construct episodes containing labelled hidden failure causes:

policy failure, (16.2)

world-model failure, (16.3)

interpretation failure, (16.4)

Purpose failure, (16.5)

Declaration failure. (16.6)

Compare:

Flat Reviser (16.7)

against:

Hierarchical Reviser. (16.8)

16.3 Primary Metric

Revision-Level Accuracy:

RLA = correct revision-level selections / all revision challenges. (16.9)

Secondary metrics:

unnecessary high-level revision,
recovery time,
catastrophic Purpose change,
catastrophic world-model replacement,
task loss.

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17. Experiment E6 — Meta-Declaration

17.1 Research Question

Can an agent detect when the representation itself should change?

Ordinary learning:

$\theta_n \to \theta_{n+1}$. (17.1)

Meta-Declaration:

$D_n \to D_{n+1}$. (17.2)

The source explicitly distinguishes ordinary parameter learning from PORE-style revision of representation or model class. $\mathbb{O} \to G_2_SO(4) \to \mathbb{H} \to \mathbb{C}^2$ 成界過程初探 1...

17.2 Candidate Tasks

Linear → nonlinear dynamics. (17.3)

Fixed environment → strategic adversary. (17.4)

Two-category ontology → three-category ontology. (17.5)

Single-level causal model → hierarchical causal model. (17.6)

17.3 Metrics

Declaration Escape Time:

T_D = steps from representational failure to successful declaration change. (17.7)

Premature Redeclaration Rate:

PR_D = unnecessary declaration changes / stable-regime opportunities. (17.8)

Structural Recovery:

SR = performance after redeclaration − performance before redeclaration. (17.9)

Declaration Cost:

C_D = switching cost + retraining cost + trace-migration cost. (17.10)

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18. Experiment E7 — Human–AI Research Dyad

18.1 Rationale

The source proposes using long-horizon Human–AI collaboration as an AGI requirements probe.

The central question is:

> Which functions does the human repeatedly supply that the AI does not yet reliably maintain on its own?

Candidate functions identified in the source include:

Persistent Purpose,
problem selection,
anomaly sensing,
reframing,
epistemic-status governance,
theory genealogy,
negative-result interpretation,
judgement of which level should be revised. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1...

18.2 Bootstrapping Method

Observe human scaffolding. (18.1)

Formalize missing function. (18.2)

Implement function in agent architecture. (18.3)

Ablate human intervention. (18.4)

Measure residual performance gap. (18.5)

Identify next missing function. (18.6)

This creates an iterative AGI requirements-discovery process.

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19. E7 Experimental Conditions

Compare:

Human + AI unrestricted collaboration. (19.1)

AI + persistent Purpose architecture. (19.2)

AI + ordinary goal prompt. (19.3)

AI + memory only. (19.4)

AI + self-revision without Purpose. (19.5)

Primary question:

Can machine-readable Purpose-management replace some human long-horizon scaffolding without reducing research coherence? (19.6)

The source directly proposes externalizing human Purpose-management into a machine-readable Purpose Belt and testing whether AI can maintain a coherent research programme for longer horizons. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1...

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20. E7 Metrics

Human Intervention Density:

HID = human interventions / research episodes. (20.1)

Research Coherence:

RC = retained valid constraints / total critical constraints. (20.2)

Silent Assumption Drift:

SAD = unannounced assumption changes / major reasoning transitions. (20.3)

Negative-Result Preservation:

NRP = retained negative results / discovered negative results. (20.4)

Theory-Genealogy Integrity:

TGI = correctly preserved theory transitions / audited transitions. (20.5)

The strongest result would be:

HID decreases while RC, NRP, and TGI do not decline. (20.6)

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21. Geometry Entrance Gate

The programme does not proceed automatically from successful Purpose engineering to complex or symplectic geometry.

A specific entrance condition is required.

Let:

U_T = Trace / observation update. (21.1)

U_P = Purpose interpretation / revision update. (21.2)

Test:

$U_T U_P \stackrel{?}{=} U_P U_T$. (21.3)

If:

$U_T U_P \approx U_P U_T$ (21.4)

across relevant regimes, then the proposed conjugate or symplectic interpretation loses its primary motivation.

If:

$[U_T, U_P] = U_T U_P - U_P U_T \neq 0$ (21.5)

robustly and functionally, then deeper geometry becomes experimentally motivated.

The source explicitly proposes this ordering and warns that complex geometry must emerge from the minimal Purpose–Observer architecture rather than be used to justify it retrospectively. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

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22. Experiment E8 — Noncommuting Updates

22.1 Procedure

For matched initial state Ω:

Condition TP:

Ω_TP = U_P(U_T(Ω)). (22.1)

Condition PT:

Ω_PT = U_T(U_P(Ω)). (22.2)

Define commutator discrepancy:

Δ_comm = d(Ω_TP,Ω_PT). (22.3)

22.2 Null Hypothesis

H0: Δ_comm ≈ 0. (22.4)

22.3 Alternative Hypothesis

H1: Δ_comm > 0 systematically. (22.5)

But nonzero difference alone is insufficient.

The noncommutation must also be:

reproducible,
functionally relevant,
stable across task variations,
not removable by trivial reparameterization.

Only then should it motivate antisymmetric structure.

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23. Experiment E9 — Purpose Geometry

If E8 succeeds, investigate whether an antisymmetric form ω_P and positive metric g_P can be independently estimated.

Define:

$A_P = g_P^{-1}\omega_P.$ (23.1)

If:

$-A_P^2 > 0,$ (23.2)

define:

$S_P = (-A_P^2)^{1/2}.$ (23.3)

Then:

$J_P = A_P S_P^{-1}.$ (23.4)

Test:

$J_P^2 \approx -I.$ (23.5)

The important experimental question is not only whether equation (23.5) can be constructed.

It is whether this representation improves:

compression,
prediction,
revision stability,
sample efficiency,
or long-horizon agency.

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24. Complex Representation Utility Test

Compare matched representations.

Real model:

$x \in \mathbb{R}^{2n}.$ (24.1)

Complex model:

$z \in \mathbb{C}^n.$ (24.2)

Keep parameter capacity and compute approximately matched.

Test:

$\text{TaskPerformance}_{\text{complex}} - \text{TaskPerformance}_{\text{real}}.$ (24.3)

$\text{Compression}_{\text{complex}} - \text{Compression}_{\text{real}}.$ (24.4)

$\text{RevisionStability}_{\text{complex}} - \text{RevisionStability}_{\text{real}}.$ (24.5)

If the complex representation is merely notationally different:

complexification has no demonstrated engineering necessity. (24.6)

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25. Experiment E10 — Structural Declaration Geometry

Only after the functional programme has survived does the programme investigate the optional Structural Declaration geometry.

The mathematical candidate is:

$A \in G_2/SO(4)$. (25.1)

with:

$A \mapsto \mathbb{H}_A$. (25.2)

The first experiment should be purely mathematical.

No AI interpretation is required.

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26. E10 Phase A — Synthetic Geodesic Recovery

Generate a starting declaration:

A_0. (26.1)

Choose hidden tangent generator:

X^*. (26.2)

Construct:

$A_1 = \exp(X^*)A_0$. (26.3)

Hide X^*.

The solver receives only:

(A_0, A_1). (26.4)

It must recover $\hat{X}$ satisfying:

$\exp(\hat{X})A_0 \approx A_1$. (26.5)

and preferably:

$\|\hat{X}\| \approx \min_X \|X\|$. (26.6)

Only after this pure mathematical solver is reliable should semantic or agent-level interpretations be introduced.

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27. E10 Phase B — Declaration Transition Prediction

If declaration states can be mapped into a geometric moduli space, test whether geometric distance predicts:

adaptation cost,
representation migration cost,
transfer difficulty,
catastrophic reset probability,
or revision time.

Candidate hypothesis:

$d_D(D_1,D_2)$ correlates with transition cost. (27.1)

If no predictive relation appears:

$G_2/SO(4)$ remains an optional mathematical analogy. (27.2)

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28. Experiment E11 — Structural vs Purpose Residual

Under the optional coupled geometry, distinguish:

R_S = Structural residual. (28.1)

R_P = Purpose residual. (28.2)

Construct three ground-truth conditions.

Structural mismatch only:

$R_S > 0, R_P \approx 0$. (28.3)

Purpose mismatch only:

$R_S \approx 0, R_P > 0$. (28.4)

Combined mismatch:

$R_S > 0, R_P > 0$. (28.5)

If the proposed decomposition cannot distinguish these conditions, the two-residual model should be rejected or simplified.

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29. Multi-Timescale Revision Experiment

The optional hierarchy proposes:

t = state dynamics, (29.1)

τ_P = Purpose revision timescale, (29.2)

τ_D = Declaration revision timescale. (29.3)

A common working hypothesis is:

$$t \ll \tau_P \ll \tau_D. \quad (29.4)$$

This should be tested rather than assumed.

Sweep:

$$\kappa_{state}, \kappa_{Purpose}, \kappa_{Declaration}. \quad (29.5)$$

Measure:

task loss,
identity stability,
adaptation speed,
false high-level revision.

If optimal systems do not exhibit higher revision resistance at higher structural levels, the proposed timescale hierarchy loses support.

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30. Capacity Matching

Ablation experiments are invalid if treatment systems simply receive more resources.

Conditions should therefore be matched on:

base model,
model version,
context budget,
inference calls,
tool access,
training data,
persistent-state budget,
maximum compute.

If the Purpose condition uses fewer named fields, unused budget should remain available to baselines as generic memory where appropriate.

The goal is to test structure, not capacity.

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31. Complexity Penalty

Let total evaluation cost be:

$$C_{total} = L_{task} + \lambda_1 C_{memory} + \lambda_2 C_{compute} + \lambda_3 C_{state} + \lambda_4 C_{revision}. \quad (31.1)$$

A larger architecture is justified only if functional improvement exceeds complexity cost.

A generic criterion is:

$$\Delta \text{Utility} > \lambda \Delta \text{Complexity}. \quad (31.2)$$

The exact λ values are application-dependent and should be preregistered in confirmatory studies.

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32. Statistical Discipline

The following standards apply to confirmatory studies.

32.1 Independent Seeds

Use multiple independently generated environments.

32.2 Paired Evaluation

Where possible, compare architectures on identical environment seeds.

32.3 Held-Out Evaluation

Development worlds and confirmatory worlds must be separated.

32.4 Preregistered Primary Metrics

Primary metrics must be fixed before confirmatory evaluation.

32.5 Confidence Intervals

Report effect estimates together with uncertainty.

32.6 Effect Size

A statistically significant but practically negligible improvement does not establish architectural necessity.

32.7 Equivalence Testing

Whenever the strong theoretical claim is irreducibility, the study should also test whether a simpler architecture is sufficiently equivalent.

This is especially important for Purpose Belt experiments.

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33. Characteristic Failure Signatures

One of the programme's strongest methodological ideas is that ablations should produce *specific* failures rather than merely lower aggregate scores.

Examples:

–Purpose Identity → reinterpretation drift. (33.1)

–Interpretation Separation → factual–normative confusion. (33.2)

–Revision Attribution → wrong-level revision. (33.3)

–Latching → oscillation. (33.4)

–Meta-Declaration → persistent ontology mismatch. (33.5)

Distinctive failure signatures provide stronger evidence for functional decomposition than overall score alone.

The source explicitly makes this argument for Purpose Belt ablations. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1…

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34. Negative-Result Registry

Every experiment should publish negative outcomes.

A standard registry entry should contain:

Experiment ID,
Hypothesis,
Architecture,
Baseline,
Primary Metric,
Observed Effect,
Confidence Interval,
Complexity Cost,
Decision.

Allowed decisions:

Retain. (34.1)

Reduce. (34.2)

Revise. (34.3)

Downgrade. (34.4)

Reject. (34.5)

Negative results must not be removed merely because later theory becomes more elegant.

---

35. Preregistration Template

Each confirmatory experiment should begin with a structured declaration.

Experiment ID:
Version:
Date:
Research Question:
Core Claim Tested:
Primary Hypothesis:
Null Hypothesis:
Baseline:
Treatment:
Ablations:
Environment Generator:
Sample Size:
Primary Metric:
Secondary Metrics:
Success Threshold:
Failure Threshold:
Equivalence Margin:
Complexity Budget:
Allowed Hyperparameter Search:
Prohibited Post-Hoc Changes:
Predicted Characteristic Failure:
Model Version:
Code Hash:
Evaluator Protocol:
Final Decision Rule:

This template is intended to prevent theory revision after results are known.

---

36. Experimental Dependency Graph

The programme should proceed approximately in the following order.

Persistent Observer Kernel
↓
Dual Residual
↓
Latching
↓

Purpose Belt
↓
Revision Hierarchy
↓
Meta-Declaration
↓
Human–AI Research Dyad
↓
Geometry Entrance Gate
↓
Noncommuting Updates
↓
Purpose Geometry
↓
Complexification
↓
Structural Declaration Geometry. (36.1)

The key constraint is:

E1–E7 do not require complex, quaternionic, or octonionic structure. (36.2)

Only after these experiments establish functional structure should deeper geometry become a primary empirical target.

---

37. Priority Order

If research resources are limited, the recommended priority is:

Priority 1 — Purpose Belt Ablation

Reason:

It has clear component-level predictions and a strong falsification condition.

Priority 2 — Revision Attribution

Reason:

Correctly identifying *what should change* may be more important than generic self-revision.

The source explicitly identifies this as a central functional addition. $\mathbb{O} \to G_2_SO(4) \to \mathbb{H} \to \mathbb{C}^2$ 成界過程初探 1...

Priority 3 — Latching and Dual Residual

Reason:

Both are simple enough for controlled synthetic tests and directly relevant to stable self-revision.

Priority 4 — Meta-Declaration

Reason:

This tests the crucial distinction between learning within a representation and changing the representation itself.

Priority 5 — Human–AI Research Dyad

Reason:

It can reveal missing long-horizon functions in current AI systems and supply requirements for later architecture.

Priority 6 — Purpose Geometry

Reason:

It should enter only after functional Purpose structure has empirical support.

Priority 7 — G2/SO(4) Declaration Geometry

Reason:

This should begin with pure mathematical recovery tests before any broader interpretation.

---

38. First Publishable Empirical Study

The strongest first confirmatory paper is likely:

Persistent Purpose and Long-Horizon Self-Revision

Ablation Tests of Purpose Identity, Interpretation, Revision Attribution, and Hierarchical Latching

The main comparison should include:

Goal Agent. (38.1)

Goal + Memory. (38.2)

Self-Revising Agent. (38.3)

Strong Conventional Meta-Learning Baseline. (38.4)

Full Purpose Architecture. (38.5)

Purpose Identity Ablation. (38.6)

Interpretation Ablation. (38.7)

Attribution Ablation. (38.8)

Latching Ablation. (38.9)

The experiment should emphasize long-horizon ontology shift rather than short-task performance.

The source explicitly states that the Purpose architecture should be tested in the combined regime of ontology shift, long horizon, value ambiguity, conflicting evidence, and self-revision. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1...

---

39. Strongest Positive Outcome

The strongest result is not:

Purpose Agent scores higher. (39.1)

It is:

different ablations generate different preregistered characteristic failures. (39.2)

For example:

–Purpose Identity → slow drift. (39.3)

–Interpretation → factual–normative confusion. (39.4)

–Attribution → wrong revision level. (39.5)

–Latching → oscillation. (39.6)

This would support the claim that the proposed decomposition is functionally meaningful rather than merely descriptive.

---

40. Strongest Negative Outcome

The cleanest negative result is:

Simple Baseline ≈ Full Architecture (40.1)

across:

action behaviour,

revision behaviour,
Purpose retention,
ontology-shift recovery,
noise resistance,
long-horizon coherence,

while using equal or lower complexity.

Then:

the strong architectural claim fails. (40.2)

The source explicitly permits this outcome: if conventional constitution, memory, meta-learning, or ordinary utility/world-model systems reproduce the same functionality without loss, Purpose Belt contributes no distinct architecture and should be treated as a re-description. 𝕆 → G2_SO(4) → ℍ → ℂ² 成界過程初探 1...

---

41. Programme-Level Decision Rules

The experimental programme should distinguish four outcomes.

Retain

The proposed component survives ablation and produces a meaningful, reproducible effect.

Reduce

The larger architecture works, but one or more components are behaviourally redundant.

Revise

The proposed effect appears, but not in the predicted form.

Reject

A simpler matched architecture reproduces the relevant behaviour or the proposed component consistently fails its preregistered function.

---

42. Relationship to Mathematical Extensions

Successful functional experiments do not automatically confirm any particular deeper geometry.

For example:

Purpose architecture success ⇏ symplectic geometry. (42.1)

Noncommuting updates ⇏ unique ω_P. (42.2)

ω_P + g_P ⇏ quaternionic compatibility. (42.3)

Quaternionic compatibility ⇏ octonionic ontology. (42.4)

Each arrow requires its own derivation or test.

This preserves the programme's central methodological discipline.

---

43. Relationship to Comparative Interpretation

Historical or philosophical comparison should occur only after independent derivation.

The experimental programme therefore does not use traditional symbolic systems to define:

benchmark categories,
revision classes,
state dimensionality,
phase count,
or expected mathematical structure.

Comparative interpretation may follow a validated result.

It may not determine one in advance.

---

44. Programme-Level Falsification

The World-Formation programme should be considered substantially weakened if repeated experiments show that:

ordinary memory reproduces Trace and Filtration effects; (44.1)

generic prediction error replaces explicit Residual; (44.2)

zero-cost revision performs as well as Latching; (44.3)

ordinary utility reproduces persistent Purpose; (44.4)

generic self-update replaces Revision Attribution; (44.5)

ordinary model selection reproduces Meta-Declaration; (44.6)

trace and Purpose updates commute; (44.7)

complex representations add no value; (44.8)

declaration geometry adds no predictive structure. (44.9)

In that case, the programme should shrink rather than protect itself through reinterpretation.

---

45. Research Standard

Every proposed extension should answer five questions.

What new distinction does it introduce? (45.1)

What prediction depends on it? (45.2)

What characteristic failure appears when it is removed? (45.3)

What simpler architecture could replace it? (45.4)

What result would cause it to be abandoned? (45.5)

If these questions cannot be answered, the extension remains exploratory.

---

46. Experimental Thesis

The entire programme can be compressed into one rule:

> **World-Formation Theory earns scientific content only when its internal arrows survive independent failure tests.**

The experimental chain is:

Formalize → Implement → Benchmark → Ablate → Falsify → Revise. (46.1)

The desired outcome is not maximal confirmation.

It is progressive reduction toward structures that cannot easily be removed.

---

47. Conclusion

The World-Formation Experimental Programme v1.0 begins from a deliberately conservative position.

It does not assume that the full theory is correct.

It does not assume that Purpose Belt is irreducible.

It does not assume that complex geometry is necessary.

It does not assume that quaternionic or octonionic structures describe AI, cognition, or physical reality.

Instead, it establishes a sequence of increasingly demanding experimental gates.

First ask whether the observer kernel matters.

Then ask whether Residual and Latching matter.

Then ask whether Purpose adds anything beyond goal, memory, and generic self-revision.

Then ask whether the system benefits from distinguishing what level should be revised.

Then ask whether it can revise its own world representation.

Only after these questions produce robust positive structure should deeper geometry enter.

The governing rules are therefore:

> **Architecture must earn its complexity.**

> **Geometry must earn its necessity.**

> **Every important arrow must be allowed to fail.**

And the central experimental principle remains:

> **Do not test the whole theory. Test the arrows.**

The next document in the series is **Preregistered Study E4: Purpose Belt Ablation**, which turns the strongest near-term claim of the programme into a concrete confirmatory protocol.

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This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.

I am merely a midwife of knowledge.